

Презентация по лабораторной работе 7

Адресация IPv4 и IPv6. Настройка DHCP

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1. Информация
2. Вводная часть
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4. Выводы

Раздел 1

1. Информация

1.1 Докладчик

► Седохин Даниил Алексеевич

1.1 Докладчик

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Раздел 2

2. Вводная часть

2.1 Цели и задачи

- ▶ Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Раздел 3

3. Выполнение заданий

3.1 слайд 1

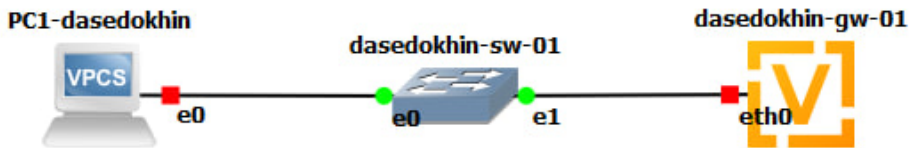
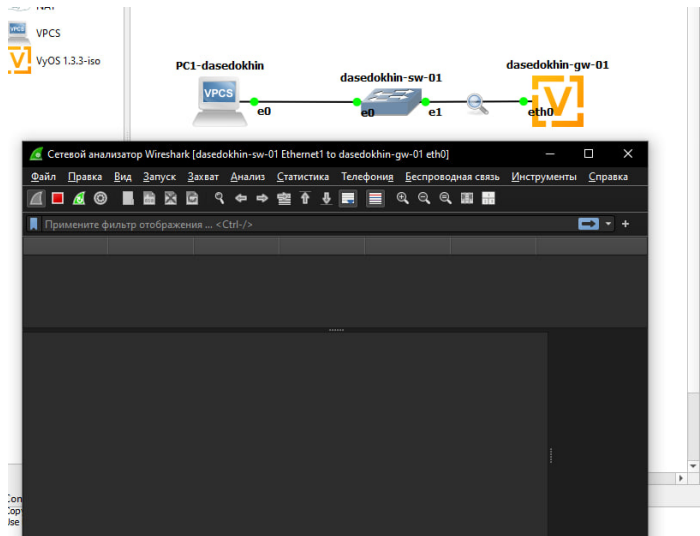


Рисунок 1: Запуск GNS3 VM и GNS3, воспроизведение топологии сети

3.2 слайд 2



3.3 слайд 3

```
dasedokhin-gw-01 - PuTTY
with partition type 83 (Linux).

Looking for appropriate partitions: (fdisk: cannot open /dev/sr: No such file or
directory
E
No suitable partition sizes found. Exiting...
vyos@vyos:~$ configure
WARNING: You are currently configuring a live-ISO environment, changes will not
persist until installed
[edit]
vyos@vyos# set system host-name dasedokhin-gw-01
[edit]
vyos@vyos# set system domain-name dasedokhin.net
[edit]
vyos@vyos# set system login user dasedokhin authentication plaintext-password 12
3456
[edit]
vyos@vyos# commit

[edit]
vyos@vyos#
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout

Welcome to VyOS - dasedokhin-gw-01 ttyS0

dasedokhin-gw-01 login: dasedokhin
Password:
Welcome to VyOS!

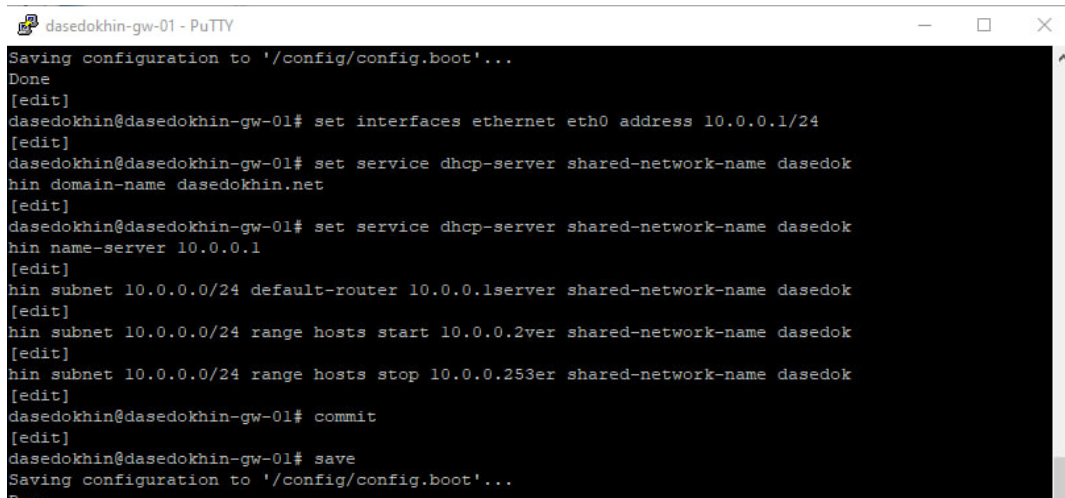
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
dasedokhin@dasedokhin-gw-01:~$ configure
WARNING: You are currently configuring a live-ISO environment, changes will not
persist until installed
[edit]
dasedokhin@dasedokhin-gw-01# delete system login user vyos
[edit]
dasedokhin@dasedokhin-gw-01# commit

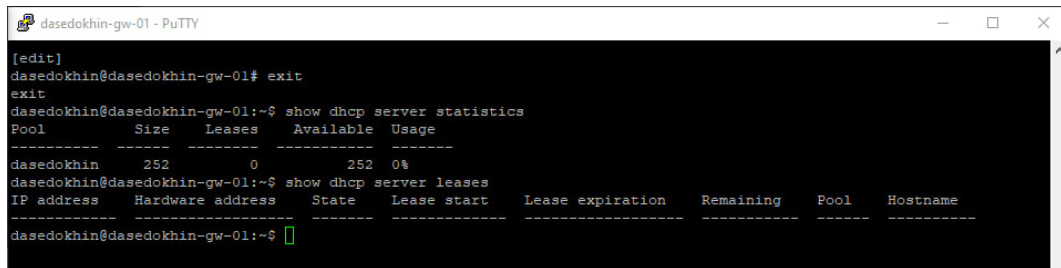
[edit]
dasedokhin@dasedokhin-gw-01#
```

3.4 слайд 4



```
dasedokhin-gw-01 - PuTTY
Saving configuration to '/config/config.boot'...
Done
[edit]
dasedokhin@dasedokhin-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
dasedokhin@dasedokhin-gw-01# set service dhcp-server shared-network-name dasedokhin domain-name dasedokhin.net
[edit]
dasedokhin@dasedokhin-gw-01# set service dhcp-server shared-network-name dasedokhin name-server 10.0.0.1
[edit]
dasedokhin@dasedokhin-gw-01# set service dhcp-server shared-network-name dasedokhin subnet 10.0.0.0/24 default-router 10.0.0.1server shared-network-name dasedokhin
[edit]
dasedokhin@dasedokhin-gw-01# set service dhcp-server shared-network-name dasedokhin subnet 10.0.0.0/24 range hosts start 10.0.0.2ver shared-network-name dasedokhin
[edit]
dasedokhin@dasedokhin-gw-01# set service dhcp-server shared-network-name dasedokhin subnet 10.0.0.0/24 range hosts stop 10.0.0.253er shared-network-name dasedokhin
[edit]
dasedokhin@dasedokhin-gw-01# commit
[edit]
dasedokhin@dasedokhin-gw-01# save
Saving configuration to '/config/config.boot'...
```

3.5 слайд 5

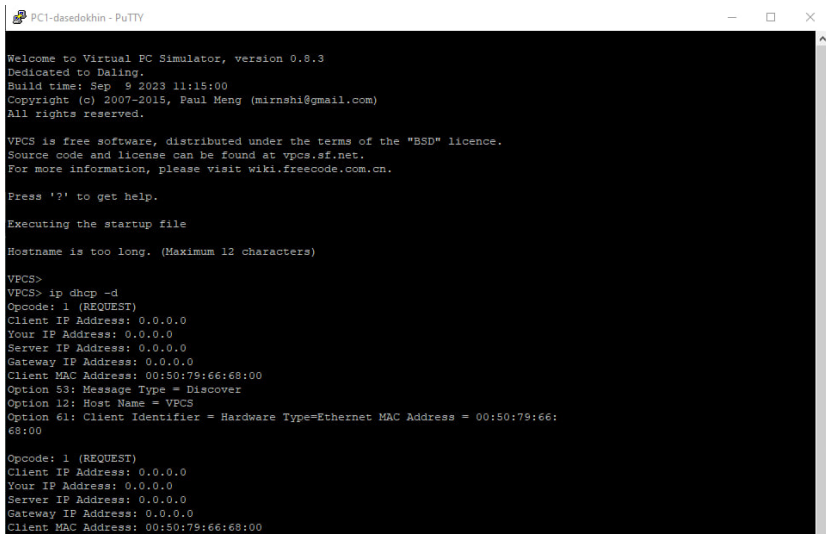


The screenshot shows a PuTTY terminal window titled "dasedokhin-gw-01 - PuTTY". The terminal output is as follows:

```
[edit]
dasedokhin@dasedokhin-gw-01# exit
exit
dasedokhin@dasedokhin-gw-01:~$ show dhcp server statistics
Pool          Size    Leases    Available    Usage
-----
dasedokhin    252      0         252         0%
dasedokhin@dasedokhin-gw-01:~$ show dhcp server leases
IP address    Hardware address    State    Lease start    Lease expiration    Remaining    Pool    Hostname
-----
dasedokhin@dasedokhin-gw-01:~$
```

Рисунок 5: Просмотр статистики DHCP-сервера и выданных адресов

3.6 слайд 6



```
PC1-dasedokhin - PuTTY

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS>
VPCS> ip dhcp -d
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00
68:00

Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
```

3.7 слайд 7

```
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> █
```

Рисунок 7: Сохранение изменений

3.8 слайд 8

```
VPCS> show ip
```

```
NAME       : VPCS[1]  
IP/MASK    : 10.0.0.2/24  
GATEWAY    : 10.0.0.1  
DNS        : 10.0.0.1  
DHCP SERVER : 10.0.0.1  
DHCP LEASE  : 86350, 86400/43200/75600  
DOMAIN NAME : dasedokhin.net  
MAC        : 00:50:79:66:68:00  
LPORT      : 20004  
RHOST:PORT  : 127.0.0.1:20005  
MTU        : 1500
```

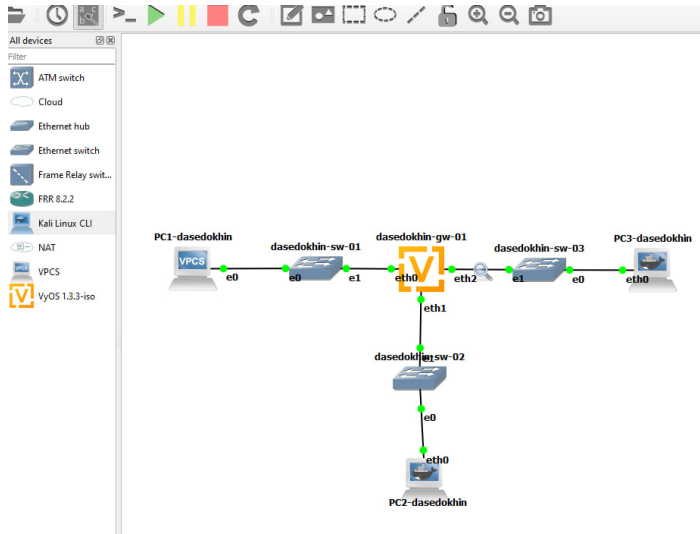
```
VPCS> ping 10.0.0.1 -c 2
```

```
84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=1.550 ms  
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=1.410 ms
```

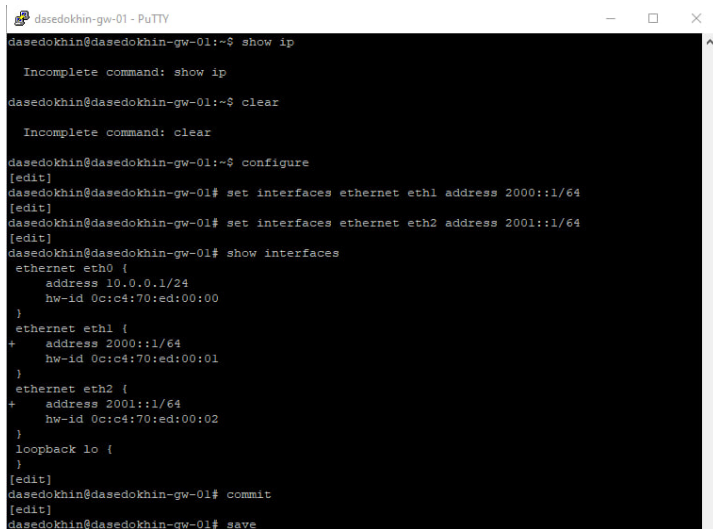
3.9 слайд 9

```
dasedokhin@dasedokhin-gw-01:~$ show log | grep dhcp
Dec 06 13:29:31 sudo[2087]:      root : TTY=unknown ; PWD=/ ; USER=root ; COMMAND=/usr/bin/sh -c /usr/sbin/vysshim /usr
/libexec/vyos/conf_mode/dhcp_server.py
Dec 06 13:29:31 vyos-configd[684]: Received message: {"type": "node", "data": "/usr/libexec/vyos/conf_mode/dhcp_serve
r.py"}
Dec 06 13:29:34 dhcpd[2108]: Wrote 0 leases to leases file.
Dec 06 13:29:34 dhcpd[2108]: Lease file test successful, removing temp lease file: /config/dhcpd.leases.1765027774
Dec 06 13:29:35 dhcpd[2110]: Wrote 0 leases to leases file.
Dec 06 13:29:35 dhcpd[2110]:
Dec 06 13:29:35 dhcpd[2110]: No subnet declaration for eth2 (no IPv4 addresses).
Dec 06 13:29:35 dhcpd[2110]: ** Ignoring requests on eth2.  If this is not what
Dec 06 13:29:35 dhcpd[2110]:   you want, please write a subnet declaration
Dec 06 13:29:35 dhcpd[2110]:   in your dhcpd.conf file for the network segment
Dec 06 13:29:35 dhcpd[2110]:   to which interface eth2 is attached. **
Dec 06 13:29:35 dhcpd[2110]:
Dec 06 13:29:35 dhcpd[2110]:
Dec 06 13:29:35 dhcpd[2110]: No subnet declaration for eth1 (no IPv4 addresses).
Dec 06 13:29:35 dhcpd[2110]: ** Ignoring requests on eth1.  If this is not what
Dec 06 13:29:35 dhcpd[2110]:   you want, please write a subnet declaration
Dec 06 13:29:35 dhcpd[2110]:   in your dhcpd.conf file for the network segment
Dec 06 13:29:35 dhcpd[2110]:   to which interface eth1 is attached. **
Dec 06 13:29:35 dhcpd[2110]:
Dec 06 13:29:35 dhcpd[2110]: Server starting service.
Dec 06 13:43:22 sudo[3105]: dasedokhin : TTY=ttys0 ; PWD=/home/dasedokhin ; USER=root ; COMMAND=/usr/libexec/vyos/op_
mode/show_dhcp.py --statistics
Dec 06 13:44:00 sudo[3131]: dasedokhin : TTY=ttys0 ; PWD=/home/dasedokhin ; USER=root ; COMMAND=/usr/libexec/vyos/op_
mode/show_dhcp.py --leases
Dec 06 13:50:02 dhcpd[2110]: DHCPDISCOVER from 00:50:79:66:68:00 via eth0
Dec 06 13:50:03 dhcpd[2110]: DHCPOFFER on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0
Dec 06 13:50:06 dhcpd[2110]: DHCPREQUEST for 10.0.0.2 (10.0.0.1) from 00:50:79:66:68:00 (VPCS) via eth0
```

3.10 слайд 10



3.11 слайд 11



```
dasedokhin-gw-01 - PuTTY
dasedokhin@dasedokhin-gw-01:~$ show ip

Incomplete command: show ip

dasedokhin@dasedokhin-gw-01:~$ clear

Incomplete command: clear

dasedokhin@dasedokhin-gw-01:~$ configure
[edit]
dasedokhin@dasedokhin-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
dasedokhin@dasedokhin-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
dasedokhin@dasedokhin-gw-01# show interfaces
  ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:c4:70:ed:00:00
  }
  ethernet eth1 {
+   address 2000::1/64
    hw-id 0c:c4:70:ed:00:01
  }
  ethernet eth2 {
+   address 2001::1/64
    hw-id 0c:c4:70:ed:00:02
  }
  loopback lo {
  }
[edit]
dasedokhin@dasedokhin-gw-01# commit
[edit]
dasedokhin@dasedokhin-gw-01# save
```

3.12 слайд 12

```
[edit]
0::/64khin@dasedokhin-gw-01# set service router-advert interface eth1 prefix 200
[edit]
ig-flaghin@dasedokhin-gw-01# set service router-advert interface eth1 other-conf
[edit]
okhin-statelessdokhin-gw-01# set service dhcpv6-server shared-network-name dased
[edit]
okhin-stateless subnet 2000::0/64service dhcpv6-server shared-network-name dased
[edit]
okhin-stateless common-options domain-search dasedokhin.net
[edit]
dasedokhin@dasedokhin-gw-01# commit
[edit]
dasedokhin@dasedokhin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
dasedokhin@dasedokhin-gw-01# run show configuration
interfaces {
    ethernet eth0 {
        address 10.0.0.1/24
        hw-id 0c:c4:70:ed:00:00
    }
    ethernet eth1 {
        address 2000::1/64
        hw-id 0c:c4:70:ed:00:01
    }
    ethernet eth2 {
        address 2001::1/64
        hw-id 0c:c4:70:ed:00:02
    }
    loopback lo {
    }
}
service {
    dhcp-server {
        shared-network-name dasedokhin {
            domain-name dasedokhin.net
```

3.13 слайд 13

root@PC2-dasedokhin: ~

(root@ PC2-dasedokhin)-[~]

ifconfig

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 2000::42:88ff:fe10:6c00 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::42:88ff:fe10:6c00 prefixlen 64 scopeid 0x20<link>
    ether 02:42:88:10:6c:00 txqueuelen 1000 (Ethernet)
    RX packets 15 bytes 1770 (1.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 15 bytes 1274 (1.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:88ff:fe10:6c01 prefixlen 64 scopeid 0x20<link>
    ether 02:42:88:10:6c:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

(root@ PC2-dasedokhin)-[~]

route -n -A inet6

Kernel IPv6 routing table

Destination	Next Hop	Flag	Met	Ref	Use	If
2000::/64	::	UAe	256	2	0	eth0
fe80::/64	::	U	256	1	0	eth0
fe80::/64	::	U	256	1	0	eth1
::/0	fe80::ec4:70ff:feed:1	UGDAe	1024	1	0	eth0
::1/128	::	Un	0	3	0	lo
2000::42:88ff:fe10:6c00/128	::	Un	0	3	0	eth0
fe80::42:88ff:fe10:6c00/128	::	Un	0	2	0	eth0

3.14 слайд 14

```
(root@ PC2-dasedokhin)~# ping 2000::1 -c 2
PING 2000::1 (2000::1) 56 data bytes
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=1.73 ms
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=1.79 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 1.730/1.757/1.785/0.027 ms

(root@ PC2-dasedokhin)~# cat /etc/resolv.conf

(root@ PC2-dasedokhin)~# dhclient -6 -S -v eth0
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on Socket/eth0
Sending on Socket/eth0
Created duid "\000\003\000\001\002B\210\0201\000".
PRC: Requesting information (INIT).
XMT: Forming Info-Request, 0 ms elapsed.
XMT: Info-Request on eth0, interval 930ms.
RCV: Reply message on eth0 from fe80::ec4:70ff:feed:1.
PRC: Done
```

3.15 слайд 15

```
(root@PC2-dasedokhin)~# ping 2000::1 -c2
PING 2000::1 (2000::1) 56 data bytes
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=1.35 ms
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=1.23 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 100lms
rtt min/avg/max/mdev = 1.233/1.293/1.353/0.060 ms

(root@PC2-dasedokhin)~# cat /etc/resolv.conf
search dasedokhin.net.

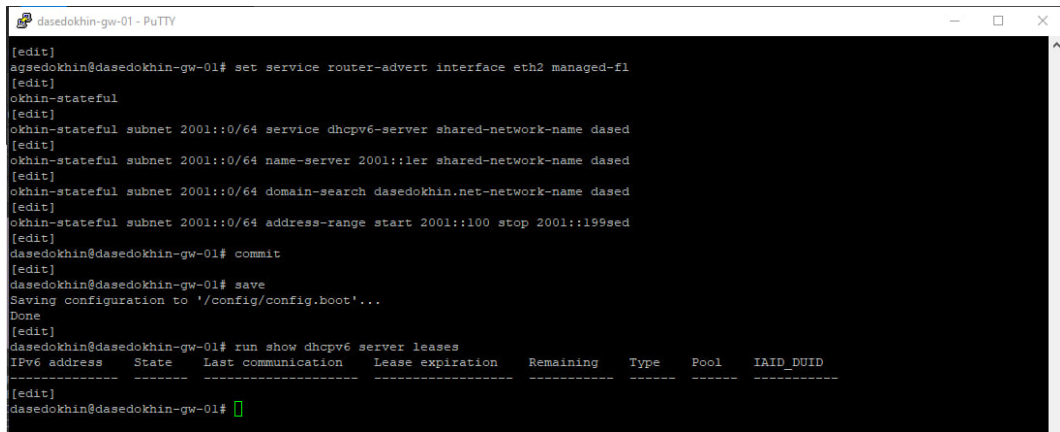
(root@PC2-dasedokhin)~#
```

3.16 слайд 16

```
dasedokhin@dasedokhin-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
[edit]
dasedokhin@dasedokhin-gw-01#
```

Рисунок 16: Просмотр статистики DHCP-сервера и выданные адреса

3.17 слайд 17



```
[edit]
agedokhin@dasedokhin-gw-01# set service router-advert interface eth2 managed-fl
[edit]
okhin-stateful
[edit]
okhin-stateful subnet 2001::0/64 service dhcpv6-server shared-network-name dased
[edit]
okhin-stateful subnet 2001::0/64 name-server 2001::ler shared-network-name dased
[edit]
okhin-stateful subnet 2001::0/64 domain-search dasedokhin.net-network-name dased
[edit]
okhin-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199sed
[edit]
dasedokhin@dasedokhin-gw-01# commit
[edit]
dasedokhin@dasedokhin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
dasedokhin@dasedokhin-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining      Type      Pool      IAID_DUID
-----
[edit]
dasedokhin@dasedokhin-gw-01#
```

Рисунок 17: Настройка DHCPv6 с отслеживанием состояния, просмотр статистики

DHCP сервера и выданные адреса

3.18 слайд 18

```
root@PC3-dasedokhin: ~  
PC3-dasedokhin console is now available... Press RETURN to get started.  
(root@ PC3-dasedokhin)~[~]  
# ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::42:67ff:fe39:8b00 prefixlen 64 scopeid 0x20<link>  
    ether 02:42:67:39:8b:00 txqueuelen 1000 (Ethernet)  
    RX packets 13 bytes 1470 (1.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 13 bytes 1006 (1006.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::42:67ff:fe39:8b01 prefixlen 64 scopeid 0x20<link>  
    ether 02:42:67:39:8b:01 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(root@ PC3-dasedokhin)~[~]  
# route -n -A inet6  
Kernel IPv6 routing table  
Destination          Next Hop              Flag Met Ref  Use If  
fe80::/64             ::                    U    256 1      0 eth0  
fe80::/64             ::                    U    256 1      0 eth1  
::/0                  fe80::ec4:70ff:feed:2 UGD Ae 1024 1      0 eth0  
::1/128              ::                    Un    0 2      0 lo  
fe80::42:67ff:fe39:8b00/128 ::                    Un    0 2      0 eth0  
fe80::42:67ff:fe39:8b01/128 ::                    Un    0 3      0 eth1  
ff00::/8              ::                    U    256 3      0 eth0  
ff00::/8              ::                    U    256 1      0 eth1  
::/0                  ::                    Un    -1 1      0 lo
```

3.19 слайд 19

```
(root@PC3-dasedokhin)~#  
$ dhclient -6 -v eth0  
Internet Systems Consortium DHCP Client 4.4.3-P1  
Copyright 2004-2022 Internet Systems Consortium.  
All rights reserved.  
For info, please visit https://www.isc.org/software/dhcp/  
  
Listening on Socket/eth0  
Sending on Socket/eth0  
Created duid "000000100000010\307\032\226\002Bg9\213\000".  
PRC: Soliciting for leases (INIT).  
XMT: Forming Solicit, 0 ms elapsed.  
XMT: X-- IA_NA 67:39:8b:00  
XMT: | X-- Request renew in +3600  
XMT: | X-- Request rebind in +5400  
XMT: Solicit on eth0, interval 1040ms.  
RCV: Advertise message on eth0 from fe80::ec4:70ff:feed:2.  
RCV: X-- IA_NA 67:39:8b:00  
RCV: | X-- starts 1765039639  
RCV: | X-- t1 - renew +0  
RCV: | X-- t2 - rebind +0  
RCV: | X-- [Options]  
RCV: | | X-- IAADDR 2001::198  
RCV: | | | X-- Preferred lifetime 27000.  
RCV: | | | X-- Max lifetime 43200.  
RCV: X-- Server ID: 00:01:00:01:30:c7:0a:86:0c:c4:70:ed:00:01  
RCV: Advertisement recorded.  
PRC: Selecting best advertised lease.  
PRC: Considering best lease.  
PRC: X-- Initial candidate 00:01:00:01:30:c7:0a:86:0c:c4:70:ed:00:01 (s: 10105, p: 0).  
XMT: Forming Request, 0 ms elapsed.  
XMT: X-- IA_NA 67:39:8b:00  
XMT: | X-- Requested renew +3600  
XMT: | X-- Requested rebind +5400  
XMT: | | X-- IAADDR 2001::198  
XMT: | | | X-- Preferred lifetime +7200  
XMT: | | | X-- Max lifetime +7500  
XMT: V IA_NA appended.  
XMT: Request on eth0, interval 970ms.  
RCV: Reply message on eth0 from fe80::ec4:70ff:feed:2.  
RCV: X-- IA_NA 67:39:8b:00  
RCV: | X-- starts 1765039640  
RCV: | X-- t1 - renew +0  
RCV: | X-- t2 - rebind +0  
RCV: | X-- [Options]  
RCV: | | X-- IAADDR 2001::198
```

3.20 слайд 20

```

root@PC3-dasedokhin ~
root@PC3-dasedokhin~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:e7ff:fe39:8b00 prefixlen 64 scopeid 0x20<link>
    inet6 2001::198 prefixlen 128 scopeid 0x0<global>
    ether 02:42:e7:39:8b:00 txqueuelen 1000 (Ethernet)
    RX packets 17 bytes 2066 (1.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 20 bytes 1758 (1.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:e7ff:fe39:8b01 prefixlen 64 scopeid 0x20<link>
    ether 02:42:e7:39:8b:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@PC3-dasedokhin~# route -n -A inet6
Kernel IPv6 routing table

```

Destination	Next Hop	Flag	Met	Ref	Use	If
2001::198/128	::	Ue	256	1	0	eth0
fe80::/64	::	U	256	1	0	eth0
fe80::/64	::	U	256	1	0	eth1
::/0	fe80::ec4:70ff:feed:2	UGDAe	1024	1	0	eth0
::1/128	::	Um	0	3	0	lo
2001::198/128	::	Um	0	2	0	eth0
fe80::42:e7ff:fe39:8b00/128	::	Um	0	3	0	eth0
fe80::42:e7ff:fe39:8b01/128	::	Um	0	3	0	eth1
ff00::/8	::	U	256	4	0	eth0
ff00::/8	::	U	256	1	0	eth1
::/0	::	!m	-1	1	0	lo

```

root@PC3-dasedokhin~# ping 2001::1 -c 2
PING 2001::1 (2001::1) 56 data bytes
64 bytes from 2001::1: icmp_seq=1 ttl=64 time=4.95 ms
64 bytes from 2001::1: icmp_seq=2 ttl=64 time=1.24 ms

--- 2001::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1000ms
rtt min/avg/max/ndev = 1.237/3.091/4.946/1.854 ms

```

3.21 слайд 21

```
[edit]
dasedokhin@dasedokhin-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication   Lease expiration      Remaining      Type           Pool                IAID_DUID
-----
2001::198         active    2025/12/06 16:47:20   2025/12/06 18:52:20   2:03:44       non-temporary   dasedokhin-stateful  00:8b:39:67:00:01
:00:01:30:c7:1a:96:02:42:67:39:8b:00
[edit]
dasedokhin@dasedokhin-gw-01#
```

Рисунок 21: Просмотр статистики DHCP-сервера и выданные адреса на маршрутизаторе

3.22 слайд 22

The screenshot shows the Wireshark interface with a packet list on the left and a packet details pane on the right. The packet list shows four packets, with the fourth packet (No. 4) selected. The packet details pane shows the structure of the selected packet, which is an ICMPv6 Router Solicitation (Type 133).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::ec4:70ff:feed::ff02::16	ff02::16	ICMPv6	170	Multicast Listener Report Message v2
2	0.279211	fe80::ec4:70ff:feed::ff02::16	ff02::16	ICMPv6	170	Multicast Listener Report Message v2
3	0.911176	::	ff02::1:ff00::1	ICMPv6	86	Neighbor Solicitation for 2001::1
4	420.600785	fe80::42:c4ff:fea2::c000	ff02::12	ICMPv6	70	Router Solicitation from 02:42:c4:a2:cd:00

Frame 4: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface -, id 0

- Ethernet II, Src: 02:42:c4:a2:cd:00 (02:42:c4:a2:cd:00), Dst: IPv6cast_02 (33:33:00:00:00:02)
 - Destination: IPv6cast_02 (33:33:00:00:00:02)
 - Source: 02:42:c4:a2:cd:00 (02:42:c4:a2:cd:00)
 - Type: IPv6 (0x86dd)
 - [Stream Index: 2]
- Internet Protocol Version 6, Src: fe80::42:c4ff:fea2:c000, Dst: ff02::12
 - 0110 Version: 6
 - 0000 0000 Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - 0000 0000 0000 0000 = Flow Label: 0x00000
 - Payload Length: 16
 - Next Header: ICMPv6 (58)
 - Hop Limit: 255
 - Source Address: fe80::42:c4ff:fea2:c000
 - Destination Address: ff02::12
 - [Stream Index: 2]
- Internet Control Message Protocol v6
 - Type: Router Solicitation (133)
 - Code: 0
 - Checksum: 0x5763 [correct]
 - Checksum Status: Good
 - Reserved: 00000000

Hex dump (0000 - 0030):

```

0000  33 33 00 00 00 02 02 42 c4 a2 cd 00 00 00 00 00 00
0010  00 00 00 10 3a ff fe 00 00 00 00 00 00 00 00 00
0020  c4 ff fe a2 cd fe ff 02 00 00 00 00 00 00 00
0030  00 00 00 00 00 02 35 00 57 63 00 00 00 00 00 00

```

↶ ↷ ↻

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, Capture, Packets, Analyze, Statistics, Telephony, Window, and Help. The toolbar contains icons for various functions like opening files, saving, and zooming. The main window is divided into three panes:

- Packet List:** Shows a list of captured packets. The first packet (No. 1) is an ICMPv6 Multicast Listener Report Message v2 from fe80::ec4:70ff:feed::ff02::1 to ff02::1:1. Subsequent packets include more reports and Router Advertisements.
- Packet Details:** Provides a hierarchical view of the selected packet (No. 118). It shows the Ethernet II frame, the IPv6 packet, and the ICMPv6 Router Advertisement message. Key fields like Arrival Time, Epoch Arrival Time, and various time deltas are displayed.
- Packet Bytes:** Shows the raw hexadecimal and ASCII representation of the packet data.

Раздел 4

4. Выводы

4.1 слайд 1

Я приобрел навыки по настройке службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.